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← Clustering Analysis

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Clustering Assignment in KNIME - Quiz Tips

Jefferson P. Carey Clustering Analysis (/learn/machinelearningwithbigdata/module/bRNLt/discussions) · 11 days ago (/learn/machinelearningwithbigdata/profiles/4e847a9aa289b44824b7d0920f9a7ec) · Edited (/learn/machinelearningwithbigdata/module/bRNLt/discussions/ANobhrSnEeWHMgod_NjFNw)

Here are the "aha" moments for me when working through this quiz. I'm interested to hear what others think of my approach, and what insights you may have. Is there an easier way to get these answers? I never discovered what to do with the 3rd output of the k-Means node. I feel like I must have missed something in the instructions or lectures.

Question 1 - You can send the k-Means node output to an Interactive Table node and browse through the results to get the answer, but you can also use the String Manipulation node to concatenate the class and cluster column values into a single value, then send the output of that node to a Value Counter node and count the occurrences of each distinct value of that concatenated string. You'll see exactly which classes were assigned to which clusters, and the number of assignments to each cluster.

Question 2 - How to handle non-numeric columns isn't addressed anywhere that I noticed, so I ignored those columns and only ran k-Means on the 3 numeric columns in the data set, and got the correct answer. There also seems to be an unexplained error unless you change the data type of the age and children columns from "Number (integer)" to "Number (double precision)" in the File Reader node, before sending that node's output to the k-Means node.

Question 3 - Again, only running k-Means on the numeric columns, the trick here is to know which of the resulting 6 clusters is the largest. Sending the k-Means node output (the 1st output) to a Value Counter node and counting the Cluster column will show you the largest cluster. Then you can also send the k-Means node output (the 2nd output) to an Interactive Table node and see the means of the largest cluster. [EDIT: See Paul Roggeveen's comment below for a better approach to answering this question.]

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question 2 is weird indeed, clustering could easily be optimized by including the other variables as well but that is not mentioned anywhere.

question 3 can simply be answered by looking in the cluster view of the K-means node where the coverage indicates the largest cluster and by clicking on it you'll find the means of the variables.

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Jefferson P. Carey · 11 days ago (/learn/machinelearningwithbigdata/module/bRNLt/discussions/ANobhrSnEeWHMgod_NjFNw/replies/hjvbTrS4EeWfHwqLjBT9w/comments/iOcwD755EeWiXxLB9mtqCw) (/learn/machinelearningwithbigdata/profiles/4e847a9aa289b44824b7d0920f9a7ec)

Wow...I'm somewhat embarrassed to say that I never realized that view was available. Seems like something I should have known. Thanks!

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KD

Kateri Drexler · 10 days ago (/learn/machinelearningwithbigdata/module/bRNLt/discussions/ANobhrSnEeWHMgod_NjFNw/replies/hjvbTrS4EeWfHwqLjBT9w/comments/sEyn3rVGeeWHMgod_NjFNw) (/learn/machinelearningwithbigdata/profiles/7c0981ebda73b7b19742fc5d8c70696)

Do you have to normalize the data before the k-means node? If so, how do you see the average of the non-normalized data?

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Jefferson P. Carey · 10 days ago (/learn/machinelearningwithbigdata/module/bRNLt/discussions/ANobhrSnEeWHMgod_NjFNw/replies/hjvbTrS4EeWfHwqLjBT9w/comments/CkwrcVTeeW2LwpPVHyhww) (/learn/machinelearningwithbigdata/profiles/4e847a9aa289b44824b7d0920f9a7ec)

I did not normalize the data. The issue was changing the data types. I also don't understand how to see the average of the non-normalized data, if the data is normalized. But I'd like to understand that, since it seems generally useful.

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FP

Edward Perry · 10 days ago

https://www.coursera.org/learn/machinelearningwithbigdata/module/bRNLt/discussions/ANobhrSnEeWHMgod_NjFNw

1/3

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 (/learn/machinelearningwithbigdata/profiles/1d442ce30855367275092dee6d4d0691) (https://tech.knime.org/getting-started (https://tech.knime.org/getting-started))

use the above link it will make your life simpler.

for question 3:

right click on the k-means module. select the clusters tables. this will give you the mean of each cluster for age, income, children.

then right click on the statistics module. select the occurrences table to find the cluster with the highest number of points.

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 Jefferson P. Carey · 10 days ago

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 (/learn/machinelearningwithbigdata/profiles/4e847a9aa289b44824bf7d0920f9a7ec)

Yes, I got that from Padri's comment above. I think the question now is, if we normalize the data before sending it to the k-Means node, how do we see the statistics for the non-normalized data? I'm assuming it's not possible, since it's not necessarily something you'd do in a real-world problem (send normalized data to k-Means but want the k-Means node to show you statistics on the non-normalized data). But it's what this quiz is asking us. The quiz doesn't explicitly suggest that we normalize the data though. Doing so was a suggestion to resolve an error resulting from integer vs. double data types that was discussed in another topic. And normalizing data before running k-Means was also discussed in the lectures.

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KD

Kateri Drexler · 10 days ago

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It was a hint in the assignment instructions.

(Hint: Clustering algorithms are based on distance and therefore normalization of the attribute values is typically required before clustering.)

I do get the correct answer by changing the data type for children and age and not using the normalizer (good job on that, btw!), but that wasn't covered anywhere in the lesson, and normalization was. So, that makes me think that it should be applied somehow. I did try the denormalization mode, and it seems like it should work, in theory. But, no correct answer with that...

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SD

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Reply

 Asim Ihsan · 10 days ago

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There are three forms of normalization supported by KNIME: min-max, z-score, and decimal scaling. Note that each scaling methods gives different k-means results. I have no idea what the course instructors want us to use, so try all three and see if you get similar answers.

- In order to use normalization your workflow looks like "File Reader" -> "Normalizer" -> "k-Means" -> "Denormalizer" (first triangle from k-Means to triangle of Denormalizer, blue square of Normalizer to blue square of Denormalizer) -> "GroupBy".
- I tried to use "Category to Number" to encode all categorical variables into a single new column, and then run k-means on this. This did not work, nor is this recommended. You're meant to do one-hot encoding, i.e. dummy variables, but I couldn't figure out how to get KNIME to do one-hot encoding.
- I then ignored all the categorical variables and only clustered using age, income, and children. Using all three forms of normalization I couldn't get the write answer for question 3.
- Then I realized my mistake, in question 3 it says "without normalization" at the end. So you need to remove the normalization / denormalization parts above and it works.

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 Asim Ihsan · 10 days ago

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Of course the first question is copied from here: https://tech.knime.org/getting-started (https://tech.knime.org/getting-started)

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AI

Amir Ilevitzky · 3 days ago

 (/learn/machinelearningwithbigdata/module/bRNLt/discussions/ANobhrSnEeWHMgod_NjFNw/replies/6xx5uLWCEeW2LwpPVHyhww/comments/rkOJ57qlEeWXjwr0-)

 (/learn/machinelearningwithbigdata/profiles/0033bd8364a99ef4bfefc7cd1642fd24)

For Question 3,

right click om the k-means module and choose "view:cluster view"

There you get for each cluster the number of items

Associated with it, and averages of age children and income for the cluster.

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BK

Bakary Konate · 2 days ago

 (/learn/machinelearningwithbigdata/module/bRNLt/discussions/ANobhrSnEeWHMgod_NjFNw/replies/6xx5uLWCEeW2LwpPVHyhww/comments/b58p6buzEeWQwoU9G12w)

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Thank you Asim for the link it's time saving.

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Duy Bui Le · 16 hours ago

(/learn/machinelearningwithbigdata/module/bRNLt/discussions/ANobhrSnEeWHMgod_NjFNw/replies/6xX5uLWCEeW2LwpPVHyhww/comments/9XVzVLysEeWXjwr0-7YQPH)

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If you don't use normalization then the k-means node will not run correctly as it is unable to parse the double precision type (income). You need to normalize that field. I couldn't know how to view the answer after normalization. The value is still in the range of 0-1

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Roger Kapp · 9 days ago

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It would help if we'd be given the answers.

So we could at least guess what we were supposed to do.

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Duy Bui Le · 16 hours ago

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Q2: I normalise the numeric data fields to a range of 0-1 and adjust the K clusters but the ID12101 is still in the same cluster_0

Q3: I normalise the data, cluster and denormalise. The largest cluster is the cluster_2 and the value of this cluster is 33.386 for age and 20,268 for income. There was one close answer to that but it was not the correct one as well.

I'm confused with my model. It is simply File Reader -> Normalise (0-1) -> K-Means -> Denormalise. The last question suggests that without normalisation but in fact this is not working. K-Means can't parse the Double type (income). We need to normalise all three of them, don't we?

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